

Printable Quiz

1. Purpose of loops

What is the main purpose of using a loop in programming?

- (A) To execute a set of statements only once
- (B) To repeat a set of statements multiple times
- (C) To compare two numbers
- (D) To define a variable

2. Loop types in Java

In Java, which of these is not a type of loop?

- (A) while
- (B) for
- (C) do while
- (D) repeat until

3. What controls loop continuation

What is the control structure that allows a loop to continue or stop?

- (A) Boolean condition
- (B) String comparison
- (C) Arithmetic operator
- (D) Input statement

4. While syntax

Which of the following is the correct general form of a while loop?

- (A) while (condition) { statements }
- (B) loop (condition) { statements }
- (C) do { condition } while { statements }
- (D) repeat (condition) { statements }

5. LCV responsibilities

In a while loop, the loop control variable (LCV) must be:

- (A) Declared but never updated
- (B) Updated only after the loop
- (C) Declared, tested, and updated inside or before the loop

- (D) Automatically managed by Java

6. Always-true while

What happens if the Boolean condition in a while loop is always true?

- (A) The loop runs once
- (B) The program crashes
- (C) The loop runs infinitely
- (D) The compiler rejects it

7. Hello World while count

How many times will this code print "Hello World"? `while (int i = 1; i < 4; i++) { System.out.println("Hello World"); }`

- (A) 3
- (B) 4
- (C) 5
- (D) Infinite

8. For header visibility

Which parts are explicitly visible in a for loop header?

- (A) Initialisation, condition, post-body action
- (B) Only the condition
- (C) Only initialisation and condition
- (D) Only post-body action

9. Infinite for

Which of the following for loop headers would create an infinite loop?

- (A) `for ( ; ; )`
- (B) `for (int i = 0; i < 5; i++)`
- (C) `for (int i = 10; i > 0; i--)`
- (D) `for (int i = 1; i <= 10; i++)`

10. For update location

In a for loop, where is the update of the LCV typically located?

- (A) Before the loop
- (B) Inside the loop body
- (C) In the third part of the loop header

- (D) After the loop ends

11. For output sequence

What output will this code produce? `for (int i = 1; i<4; i++) System.out.println("Num: " + i);`

- (A) Num: 0, Num: 1, Num: 2
- (B) Num: 1, Num: 2, Num: 3
- (C) Num: 1, Num: 2, Num: 3, Num: 4
- (D) None

12. While vs Do-while difference

What is the main difference between a while loop and a do while loop?

- (A) while runs once, do while runs many times
- (B) do while runs its body before testing the condition
- (C) do while can't use Boolean conditions
- (D) There is no difference

13. Do-while syntax

Which of the following is the correct syntax for a do while loop?

- (A) `do { statements } while (condition);`
- (B) `while { statements } do (condition);`
- (C) `loop (condition) { statements }`
- (D) `do while { statements }`

14. When to choose do-while

When should you choose a do while loop instead of a while loop?

- (A) When the body must run at least once
- (B) When the condition should never be checked
- (C) When the loop must never terminate
- (D) When you don't need a counter

15. LCV definition

What do we call a variable that controls how many times a loop executes?

- (A) Sentinel variable
- (B) Loop control variable
- (C) Incrementer
- (D) Boolean variable

Answer Key

1. B
2. D
3. A
4. A
5. C
6. C
7. B
8. A
9. A
10. C
11. B
12. B
13. A
14. A
15. B